AIRFOIL SELECTION & AERODYNAMIC ANALYSIS

Technical report | Problem 1 | RC aircraft design

1. Objective and design assumptions

Two different missions require different aerodynamic priorities. Aircraft A carries the largest practical payload while remaining stable; Aircraft B is intended for aerobatic manoeuvres, including turns, rolls and loops. This report selects airfoil sections only; the final aircraft must still be designed around wing planform, centre of gravity, tail volume, control authority, structural strength and propulsion.

Working assumptions. Low-speed, subsonic electric RC aircraft; clean, rigid wings; conventional tail; no flaps; and a representative section Reynolds number of approximately 2×10^5 for discussing low-Reynolds-number behaviour. For air at ordinary conditions, this corresponds approximately to a 0.15 m chord at 20 m/s. This is a comparison design point, not a measured operating condition for either finished aircraft.

2. Candidate airfoils

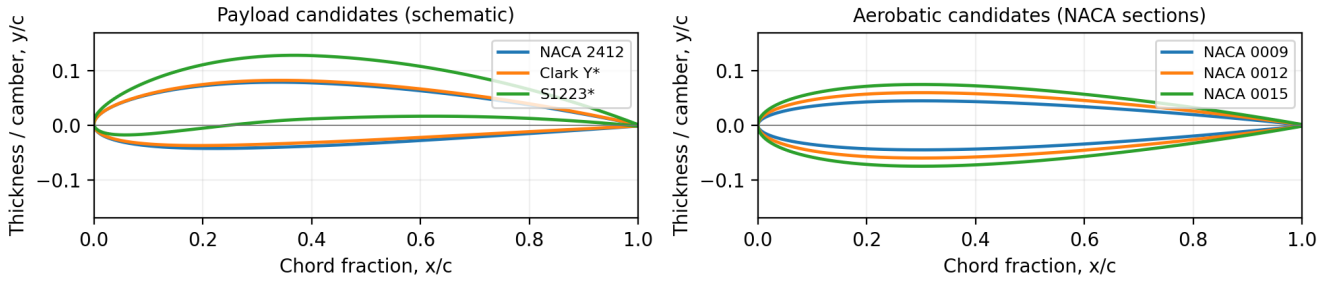
Mission	Candidate	Geometry / intended advantage	Important limitation
A — payload	**Clark Y**	Cambered, relatively thick section; practical flat-bottom construction and useful lift at modest speed.	Not a guarantee of gentle stall; surface finish and complete wing planform matter.
A — payload	**NACA 2412**	2% camber at 40% chord; 12% thickness. Moderate camber and useful structural depth.	Less specialised for very high lift than S1223; pitching moment and drag still need checking.
A — payload	**Selig S1223**	High-lift, low-Reynolds-number section; designed for heavy-lift / slow-flight applications.	Strong pitching-moment demand and potentially demanding pressure recovery; trim and stall behaviour need care.
B — aerobatic	**NACA 0009**	Symmetric; 9% thickness. Thin section can reduce profile thickness and may suit a fast, responsive wing.	Less internal depth and potentially less structural stiffness; no camber-based lift bias.
B — aerobatic	**NACA 0012**	Symmetric; 12% thickness. Balanced baseline for upright and inverted flight with practical structure.	Does not inherently provide high maximum lift; wing and control design remain decisive.
B — aerobatic	**NACA 0015**	Symmetric; 15% thickness. More depth for stiffness and internal structure.	Greater thickness can increase profile drag; not automatically more manoeuvrable.

3. Geometry and aerodynamic coefficients

The NACA four-digit designation gives maximum camber (first digit, in percent chord), its position (second digit, in tenths of chord), and maximum thickness (last two digits, in percent chord). Thus NACA 2412 has 2% camber at 40% chord and 12% thickness; NACA 0012 is symmetric and 12% thick. Symmetry means the upper and lower surfaces mirror each other about the chord line.

Lift and drag are commonly written as $L = \tfrac{1}{2}\rho V^2 S C_L$ and $D = \tfrac{1}{2}\rho V^2 S C_D$. C_L and C_D are dimensionless section/wing coefficients whose values depend on angle of attack, Reynolds number, surface condition and (for a finite wing) three-dimensional effects. L/D is a useful efficiency ratio, but maximum L/D is not the same objective as minimum sink rate or maximum payload.

Camber generally shifts the lift curve so positive lift is produced at a smaller or even negative geometric angle of attack. This helps a payload aircraft operate at useful lift coefficients, but cambered sections commonly have a non-zero pitching moment. A symmetric section has approximately zero lift at zero angle of attack and behaves more similarly in upright and inverted flight, which is desirable for aerobatics. Thickness provides room for spars and increases structural depth, but excessive thickness may raise profile drag.



*Clark Y and S1223 curves are illustrative cambered shapes, not coordinate-accurate profiles.

Figure 1. Conceptual profile comparison. NACA 4-digit profiles are generated from the standard analytic thickness/camber equations; Clark Y and S1223 outlines are illustrative only and must not be used as manufacturing coordinates.

4. Lift, drag, efficiency and stall comparison

A fair numerical comparison requires polars generated or measured at the same Reynolds number, Mach number, angle-of-attack range, transition assumption and surface roughness. The assignment does not specify aircraft chord or speed, and no XFOIL/XFLR5 run has been performed for this report. Therefore the table below is a defensible qualitative comparison, not a fabricated numerical ranking. The published S1223 result is reported separately as experimental evidence.

Section	C_L behaviour	C_D / L/D	Stall / handling implications
Clark Y	Camber gives positive lift at modest incidence.	Moderate drag; efficiency depends strongly on Re and build quality.	Often used on slow models, but stall progression is not guaranteed by section name alone.
NACA 2412	Positive lift at zero geometric incidence; moderate lift capability.	Moderate profile drag; reasonable general-purpose compromise.	Camber can make stall onset and pitching moment differ from a symmetric section.
S1223	Very high maximum lift demonstrated at $Re \approx 2 \times 10^5$.	High lift does not mean lowest drag at every C_L; high trim drag may reduce aircraft efficiency.	Needs careful trim and testing; do not assume gentle stall from high $C_{L,max}$.
NACA 0009	Near-zero lift at zero incidence; lift rises with angle of attack.	Thin section may have lower profile drag in some regimes.	Symmetric response supports inverted flight; reduced thickness can constrain structure.
NACA 0012	Near-zero lift at zero incidence; broadly symmetric lift response.	Baseline symmetric section; actual drag polar is Re- and transition-dependent.	Useful neutral starting point; finite-wing stall still depends on planform/twist.
NACA 0015	Similar symmetric lift-curve concept; greater thickness.	Thickness may increase profile drag compared with a thinner section.	Structural depth is useful, but thickness alone does not ensure better control or stall behaviour.

Published quantitative anchor — S1223. Selig and Guglielmo report a wind-tunnel maximum lift coefficient of about 2.2 for S1223 at $Re = 2 \times 10^5$ (the paper tabulates $C_{L,max} = 2.23$ and quarter-chord pitching-moment coefficient $C_{m,c/4} = -0.29$). This is a two-dimensional section result under the paper's test conditions, not a guaranteed aircraft value. The large negative pitching moment is an important design penalty: the tail may need to balance it, increasing trim drag.

For NACA 0012, NASA's critical assessment compiles data from more than 40 wind tunnels and explicitly notes experimental scatter and sensitivity to Reynolds number, transition and facility effects. This is why values from unrelated online polar plots should not be mixed into one "same-condition" comparison.

5. Stall and stability

Stall occurs when angle of attack exceeds the section/wing's critical value and flow separation causes lift to stop increasing and usually fall. It is not simply "the aircraft becoming too slow": speed, weight, wing area and $C_{L,max}$ determine the stall speed, while angle of attack triggers the stall. For a level 1-g condition, $V_s \approx \sqrt{[2W/(\rho S C_{L,max})]}$. A high $C_{L,max}$ can lower stall speed for fixed weight and wing area, but it does not by itself establish benign stall behaviour. Washout, taper, aspect ratio, surface finish, centre of gravity and tail/control authority all matter.

6. Recommended selections and trade-offs

Aircraft A — preliminary selection: Selig S1223. Its documented high-lift performance at a Reynolds number relevant to many RC models makes it the strongest candidate when maximum payload and low-speed lift dominate. The selection is conditional: the designer must check pitching-moment balance, trim drag, stall progression, structural thickness/spar placement and whether the actual aircraft Reynolds number is near the design range. If predictable handling, easier construction or a more forgiving first prototype outweighs peak lift, Clark Y or NACA 2412 may be the safer practical choice.

Aircraft B — preliminary selection: NACA 0012. A symmetric section is a sensible baseline for aerobatics because its lift behaviour is more alike in upright and inverted flight than a strongly cambered section. The 12% thickness is a practical compromise between structural depth and profile thickness. NACA 0009 is a thinner alternative if lower thickness drag and a faster design are prioritised; NACA 0015 is a thicker alternative if structural depth is more important. Neither alternative should be selected without checking the actual Reynolds-number polar and stiffness requirements.

Can one airfoil serve both missions? A symmetric section can be used on both aircraft, and a cambered section can perform aerobatics, but neither is optimal for both missions. A symmetric section generally needs more incidence or speed to generate the lift a cambered section can provide at low incidence; a high-lift cambered section can generate a large pitching moment and is less neutral for inverted flight. If cost or manufacturing simplicity forces one common section, NACA 0012 is a reasonable compromise, but Aircraft A may need greater wing area, lower wing loading or higher operating speed to carry the same payload.

7. Airfoil versus whole-aircraft performance

Airfoil data describe a two-dimensional cross-section. Actual aircraft performance also depends on: (i) wing loading W/S and wing area, which strongly affect stall speed; (ii) aspect ratio and planform, which affect induced drag; (iii) taper, twist and washout, which influence stall progression; (iv) centre of gravity, tail size and tail moment arm, which determine stability and trim; (v) aileron/elevator/rudder sizing and servo authority, which determine control response; (vi) structural stiffness and mass distribution; and (vii) motor, propeller and battery, which determine available thrust and endurance. A highly manoeuvrable aircraft is not created by choosing a symmetric airfoil alone.

8. Conclusion

S1223 is recommended as the high-lift candidate for the payload aircraft, subject to trim and stall validation. NACA 0012 is recommended as the balanced symmetric baseline for the aerobatic aircraft. The six shortlisted profiles cover the key design trade-off: cambered/high-lift sections favour lift at low speed, while symmetric sections favour similar upright and inverted behaviour. Final selection should be confirmed with matched-Reynolds-number polars and flight testing.

References

[1] M. S. Selig and J. J. Guglielmo, “High-lift low Reynolds number airfoil design,” *Journal of Aircraft*, vol. 34, no. 1, pp. 72–79, 1997. DOI: 10.2514/2.2137. Experimental S1223 data and design discussion: https://m-selig.ae.illinois.edu/pubs/GuglielmoSelig-1997-JofAC-S1223.pdf

[2] UIUC Applied Aerodynamics Group, “UIUC Airfoil Data Site / Coordinates Database,” https://m-selig.web.engr.illinois.edu/ads.html (airfoil coordinate source and low-speed airfoil resources).

[3] W. J. McCroskey, “A Critical Assessment of Wind Tunnel Results for the NACA 0012 Airfoil,” NASA TM-100019, 1987. https://ntrs.nasa.gov/citations/19880002254

[4] UIUC Low-Speed Airfoil Tests, “S1210/S1223 reference,” notes on heavy-lift / slow-flight design range and Reynolds-number context: https://m-selig.ae.illinois.edu/ads/ref/s1210s1223_ref.html

Scope and integrity note. This report includes a qualitative six-section comparison and cites published S1223 and NACA 0012 evidence. It does not claim to contain six matched numerical polars, an XFOIL/XFLR5 simulation, or exact coordinate drawings for Clark Y and S1223. Those are remaining improvements if the +5 simulation bonus is pursued.